MTH001 (Live Quiz)

Mid Term

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Question: Consider the set f={(xA2,x):x∈R}f = \{(xA2, x) :x \in \mathbb{R} \}. Is this a function from R\mathbb{R} to R\mathbb{R}?

- False ✅
- True

MCQ 2:

Question: Identify the sequence as Arithmetic, Geometric, or neither. 2,6,18,54,162,…2, 6, 18, 54, 162, \ldots

- Arithmetic
- Neither
- Geometric ✅

MCQ 3:

Question: For an arithmetic sequence with a1=1a_1 = 1, if the sum of its 101 terms is also 101, then the common difference of this A.P is:

- 11 ✅
- 101101
- 100100

MCQ 4:

Question: The option " " is true if the co-domain of f=f = range of ff.

- one-to-one ✅
- many-to-one

MCQ 5:

Question: Domain of the function y=xy = \sqrt{x} is:

- zero or negative real numbers
- only positive real numbers
- zero or positive real numbers ✅
- only negative real numbers

MCQ 6:

Question: Range of the function is always a:

- Subset ✅
- Proper Subset
- Power set

MCQ 7:

Question: Sum up the following series: 50+49+…+2150 + 49 + \ldots + 21.

- 100100
- 10651065 ✅
- 10501050
- 106106

MCQ 8:

Question: The sum of an infinite geometric series 2+1+12+…2 + 1 + \frac{1}{2} + \ldots with a=2a = 2 and r=12r = \frac{1}{2} will be:

- 1212 ✅
- 1313
- 1414

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- 1111

MCQ 9:

Question: When a1=9a_1 = 9 and a2=18a_2 = 18, the explicit formula for the sequence will be (where "A" denotes exponentiation):

- an=2×9n−1a_n = 2 \times 9^{n-1}
- an=9+2(n−1)a_n = 9 + 2(n-1)
- an=9×2n−1a_n = 9 \times 2^{n-1} ✅
- an=9×2na_n = 9 \times 2^n

MCQ 10:

Question: How many one-to-one functions are there from a set with three elements to a set with four elements?

- 4+4×3=104 + 4 \times 3 = 10
- 4×3×2=244 \times 3 \times 2 = 24 ✅
- 4×3+2=144 \times 3 + 2 = 14
- 3×4×1=133 \times 4 \times 1 = 13

. The 9999th term of the sequence: {-U-Li-1),..} is

- 9999
- **-9999 ✓**
- O
- O

2. If f is a function from A={a,b,c} to B={1,2,3} such that f={(a,3),(b,1),(c,2)}, then f is called

- Surjective function
- Bijective function ✓
- Injective function
- None of these

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3. If f is a function from A={1,2} to B={a,b,c} such that f={(1,a),(2,b)}, then f is called

- None of these
- Bijective function
- Surjective function
- **Injective function ✓**

4. f is a function from ℕ to ℕ defined as f(n) = 2n + 1. This function is injective

- **True ✓**
- False

5. The explicit formula for the sequence 15, 12, 9, 6, . . . is

- an = 15 - (n-1)3 ✓
- an = 15 * (n-1)3
- an = 15 - (n)3
- an = 15 + (n-1)3

6. How many one-to-one functions are there from a set with two elements to a set with three elements?

- **3 ✓**
- 2^(3+4)
- 2^(3 x 4)
- 2^4

7. Consider the set f = {(x^2, x): x ∈ ℝ}. Is this a function from ℝ to ℝ?

- **True ✓**
- False

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8. Suppose X is a set with three elements and Y is a set with four elements. How many binary relations are there from X to Y?

- 243 ; That is, 2 is raised to the power 3
- 2^(3+4) ; That is, 2 is raised to the power (3 + 4)
- **2^(3 x 4) ; That is, 2 is raised to the power (3 x 4) ✓**
- 2^4 ; That is, 2 is raised to the power 4

9. A function is called surjective.

- **Onto function ✓**
- None of these
- Into function
- (1-1) and onto function

10. If A is any given set, then the relation in A is the ____ of the Cartesian product A × A.

- Power Set
- **Subset ✓**
- Proper Subset

1. **Let A = {1, 2, 3} and R = {(1, 2), (2, 3), (1, 3), (2, 1)} be a relation on A. Which of the following ordered pairs are needed in R to make it transitive?**
    - (11) and (2, 2) ☐
    - (2, 2) and (3, 2) ☐
    - (11) and (3, 3) ☐
    - (3, 1) and (3, 2) ☑

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2. **A relation S on set Y is iff there is an element 'a' belonging to X such that (a, a) belongs to S.**
    - transitive ☐
    - irreflexive ☐
    - not irreflexive ☑

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3. **Which of the following is a subset of the set {1, 3, 5, 7, 9, 11, 13, 15}?**
    - {13, 6} ☐
    - {1, 9, 10, 11} ☐
    - {5, 7, 1, 15} ☑
    - {3, 7, 9, 11, 14, 15} ☐

4. **Let A = {1, 2, 3} then which of the following is a symmetric relation on A?**
    - {(1, 4), (2, 1), (1, 3), (3, 1)} ☐
    - {(1, 4), (1, 2), (2, 1), (3, 3), (3, 1)} ☐
    - {(1, 2), (1, 3)} ☐
    - {(1, 2), (2, 1), (1, 3), (3, 1)} ☑

5. **If in the directed graph of a relation R there is no pair of arrows between two distinct points of the set A, then R is**
    - antisymmetric ☑
    - symmetric ☐

6. **Consider the relation R = {(a, b), (a, c), (c, b), (b, c)} on the set A = {a, b, c}. Is R symmetric?**
    - True ☐
    - False ☑

7. **Let R be a binary relation on a set A. R is anti-symmetric if and only if**
    - For some a, b belonging to A if (a, b) belongs to R. ☐
    - For all a, b belonging to A if (b, a) belongs to R. ☐
    - For some a, b belonging to A if (a, b) belongs to R and (b, a) belongs to R then a = b. ☑
    - For all a, b belonging to A if (a, b) belongs to R and (b, a) belongs to R then a = b. ☐

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8. **Let A = {1, 2, 3} and R = {(1, 2), (2, 1), (3, 1)} be a relation on A. Which ordered pair is needed in R to make it symmetric?**
    - (3, 3) ☑
    - (1, 1) ☐
    - (1, 3) ☐

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9. **Let O = {-4, -3, -2, 0, 3, 6} and P is the set of odd numbers, then O intersection P =**
    - O ☐
    - {-4, -3, -2, 0, 3, 6} ☐
    - Set of odd numbers ☐
    - {-3, 3} ☑

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10. **Let V = {-2, 0, 4, 7, 9} and W is the set of odd numbers, then V intersection W =**

- O ☐
- {-2, 0, 4, 7, 9} ☐
- Set of odd numbers ☐
- {7, 9} ☑
- {4, 7, 9} ☐

1. **In the truth table of an argument, the critical row is the row where all ___ are true.**
    - implications ☑
    - arguments
    - conclusions
2. **Let p and q be two propositions then, ~ (pV ~ q) =**
    - ~PΛ~q
    - ~PΛq ☑
    - PVq
3. **If A = {0} which of the following statement is correct?**

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- 0∈A ✅
- a∈0

4. **Let p = "you have fever", q = "you miss the final match" and r = "you attend the ceremony". Translate the statement "If you have fever, then you will miss the final match" in logical expression.**
   - r > p
   - p > q ✅
   - ~qVr
   - q > p
5. **An argument is invalid if all the premises are true but the ___ is false.**
   - conclusion ✅
6. **What is the truth value of q V ~r when q is true and r is false?**
   - T ✅
   - F
7. **What is the truth value of ~(p ∧ q) when p is true and q is false?**
   - F
   - T ✅
8. **In which method are the elements of the set written within braces?**
   - None of these
   - Set builder
   - Descriptive
   - Tabular ✅
9. **Identify the irrational number from the following options:**
   - 0.3333.
   - π ✅
   - 0.75
   - -5
10. **Which of the following statements accurately describes the null set?**
    - The null set is a proper subset of every set.

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- The null set contains only one element, which is 0.
- The null set is a subset of every set. ✅
- The null set is equivalent to the set containing the empty string.

⍰ If p is true and q is false then what will be the truth value of ~(p@q)?

- True
- False ✓

⍰ If there are two variables p and q in a logical statement then the biconditional operator can be represented as____.

- p ⇒ q
- p ↔ q ✓
- p → q
- p ↔ q

⍰ The statement form p ∨ ~ p is____.

- tautology ✓
- falsity
- negation of p
- negation of q

⍰ Let n= "Amna is good in English grammar" and s = "Amna is a psychology student". How can we translate the statement "Amna is neither good in English grammar nor a psychology student" in logical expression.

- n ∧ s
- ~n ∧ s
- n ∧ ~s
- ~n ∧ ~s ✓

⍰ If B = {a, b} and C = {a, b, c} then which of the following statements is true?

- C ⊂ B
- B and C are disjoint sets.

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- B ⊆ C and B = C
- B ⊆ C and B ≠ C ✓

What is a truth value of p ∨ q when p is true and q is false?

- F
- T ✓

What is the negation of the statement "If n is divisible by 15, then n is divisible by 5 and n is divisible by 3"?

- n is divisible by 5 or 3 but not divisible by 15.
- n is divisible by 15 and 3 but not divisible by 5.
- n is divisible by 15 but n is not divisible by 5 or by 3. ✓
- n is divisible by 15 and 5 but not divisible by 3.

Which of the following expressions represents a set using set builder notation?

- {1,2,3,4,5}.
- The set of all vowels in the English alphabet.
- {x | x is a prime number less than 10}. ✓
- {apple, orange, banana, pineapple}

Let p and q be two statements, then the conjunction of p and q is denoted as

- P Δ q
- P ∨ q
- P ∧ q ✓
- P q

If A is a subset of B then A intersection B =

- Universal set
- A ✓
- Empty set

1. **What is a truth value of \(\neg (p \vee q)\) when \(p\) is false and \(q\) is true?**

- F

- T

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- **Correct Option: F**

2. **What will be the negation of the statement "$-3 \leq x \leq 7$" by DeMorgan's law?**

- $x < -3 \text{ or } x > 7$
- $x \leq -3 \text{ or } x \geq 7$
- $x \leq -3 \text{ or } x > 7$
- $x < -3 \text{ and } x > 7$
- **Correct Option: $x < -3 \text{ or } x > 7$**

3. **Which of the following expressions represents a set using set builder notation?**

- $\{apple, orange, banana, pineapple\}$
- $\{x \mid x \text{ is a prime number less than 10}\}$
- $\{1, 2, 3, 4, 5\}$
- $\text{The set of all vowels in the English alphabet}$
- **Correct Option: $\{x \mid x \text{ is a prime number less than 10}\}$**

4. **Consider sets $A = \{2, 3\}$ and $B = \{x \mid x \text{ is a prime number less than 4}\}$. Which of the following statements regarding sets A and B is true?**

- Sets A and B are equal.
- Sets A and B are not equal.
- Set B is a proper subset of set A.
- Set A is a proper subset of set B.
- **Correct Option: Sets A and B are equal.**

5. **Consider that q and p are statements such that $q \rightarrow p$ is false. Then the truth value of $q \leftrightarrow p$ is ___.**

- False
- True
- **Correct Option: False**

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1. Which of the following numbers is rational?

Options:

A) π

B) e

C) √20.75

Correct Option: C) √20.75

2. In which method the elements of the set are written within braces?

Options:

A) Tabular

B) None of these

C) Descriptive

D) Set builder

Correct Option: D) Set builder

3. What is the cardinality of a set?

Options:

A) The product of all elements in the set.

B) The sum of all elements in the set.

C) The largest element in the set.

D) The number of elements in the set.

Correct Option: D) The number of elements in the set.

4. {x / x belongs to set of natural numbers and 2 < x < 4} is _____.

Options:

A) Empty set

B) An infinite set

C) A finite set

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D) A singleton set

Correct Option: B) An infinite set

5. Which of the following sets is finite?

Options:

A) Set of natural numbers less than or equal to fifty.

B) Set of animals on earth.

C) Set of points in a circle.

D) Set of lines which are parallel to a given line.

Correct Option: A) Set of natural numbers less than or equal to fifty.

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